

# Supplemental Material for “Defect-aware graph learning of impurity incorporation energetics and chemical transferability in two-dimensional materials”

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## I. POST-HOC SCHNET GRAPH-READOUT SENSITIVITY

The prespecified periodic SchNet comparator follows the original energy-oriented construction [1]: its learned atomwise scalar contributions are summed to produce the graph prediction. Defect formation energy is instead localized and non-extensive with respect to supercell atom count, and prior atomistic studies show that graph pooling can materially affect localized or intensive quantum-property prediction [2]. We therefore performed one bounded post-hoc sensitivity analysis after completing the main comparison.

The intervention changed only the graph readout from atomwise addition to mean pooling. It retained all five immutable host-held-out folds, seeds 342–344, the SchNet interaction stack, learned atomic-number input, optimizer, learning rate schedule, host-balanced sampler, target normalization, target-noise stream, gradient clipping, and 150-epoch budget. Predictions were averaged over the three seeds within each fold. The pooled mean-minus-add difference in absolute error was evaluated with 20,000 bootstrap samples clustered by the 44 host materials.

TABLE I. Post-hoc SchNet graph-readout sensitivity under the fixed host-held-out protocol. Each fold prediction averages seeds 342–344. The difference is mean-readout MAE minus additive-readout MAE, in eV. The pooled host-cluster bootstrap 95% interval for this difference is [-7.704, -1.256] eV. The prespecified additive-readout result remains the main comparator.

| Partition         | $n$          | Add MAE      | Mean MAE     | Mean – add    |
|-------------------|--------------|--------------|--------------|---------------|
| Fold 1            | 2023         | 3.882        | 2.179        | -1.704        |
| Fold 2            | 2019         | 8.199        | 2.068        | -6.132        |
| Fold 3            | 2069         | 5.484        | 2.530        | -2.954        |
| Fold 4            | 2057         | 5.466        | 2.468        | -2.998        |
| Fold 5            | 2056         | 10.474       | 3.501        | -6.973        |
| <b>Pooled OOF</b> | <b>10224</b> | <b>6.703</b> | <b>2.552</b> | <b>-4.151</b> |

Mean pooling reduced the host-held-out MAE in all 5/5 folds. Across individual samples, the Spearman association between atom count and absolute error changed from 0.278 for additive pooling to 0.010 for mean pooling. Across the 44 hosts, the corresponding association between median atom count and host MAE changed from 0.312 to 0.218. These descriptive associations are consistent with a size-sensitive additive readout, but atom count covaries with host chemistry and therefore does not identify a standalone causal mechanism. The intervention establishes that readout choice contributes to the extreme additive-SchNet error; it neither replaces the prespecified main comparator nor attributes the remaining DART–SchNet difference to a single component.

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- [1] K. T. Schütt, H. E. Sauceda, P.-J. Kindermans, A. Tkatchenko, and K.-R. Müller, SchNet – a deep learning architecture for molecules and materials, *The Journal of Chemical Physics* **148**, 241722 (2018).  
[2] D. Buterez, J. P. Janet, S. J. Kiddle, D. Oglic, and P. Lið, Modelling local and general quantum mechanical properties with attention-based pooling, *Communications Chemistry* **6**, 262 (2023).